

Intro to Rational Expressions – Fractions and Exponents Review – Worksheet

MCR3U

1) Add or subtract the following fractions.

a) $\frac{2}{3} + \frac{3}{4}$

b) $\frac{3}{5} - \frac{1}{3}$

c) $\frac{x}{3} + \frac{y}{2}$

d) $\frac{3x}{5} - \frac{2y}{7}$

2) Simplify (multiplication)

a) $\left(\frac{2}{3}\right)\left(-\frac{1}{5}\right)$

b) $\left(\frac{4}{5}\right)\left(\frac{25}{12}\right)$

c) $\left(\frac{17}{8}\right)\left(\sqrt{\frac{13}{289}}\right)$

d) $\left(-\frac{5}{8}\right)\left(\sqrt{\frac{75}{128}}\right)$

3) Simplify (multiplication and division)

a) $\frac{4}{9} \cdot \frac{7}{4}$

b) $\frac{-11}{5} \cdot \frac{-7}{4}$

c) $\frac{-17}{10} \div \frac{9}{4}$

d) $\frac{6}{7} \div \frac{7}{6}$

e) $\frac{7}{3} \cdot 6$

4) Simplify (exponents)

a) $(x^6)(x^4)$

b) $(y^3)(y^7)(y^9)$

c) $m^5 \div m^4$

d) $h^6 \div h^8$

e) $\sqrt{h^6} \div \sqrt{h^8}$

f) $\frac{x^{10}}{x^5}$

g) $(yz^2)^3$

h) $\left[-(x)^2\right]^2$

i) $(x^2)^3(y^3)^2$

j) $\frac{(-x)^2}{(-x)}$

5) Simplify fractions and exponents

a) $\frac{7x}{3y} \cdot \frac{12y}{5x^2} + \frac{2}{3}$

b) $\frac{25x^3}{3x} \cdot \frac{7y^2}{5yx^2}$

6) Simplify. Your answer should contain only positive exponents.

a) $(x^{-2}x^{-3})^4$

b) $(x^4)^{-3} \cdot 2x^4$

c) $(n^3)^3 \cdot 2n^{-1}$

d) $(2v)^2 \cdot 2v^2$

e) $\frac{2x^2y^4 \cdot 4x^2y^4 \cdot 3x}{3x^{-3}y^2}$

f) $\frac{2y^3 \cdot 3xy^3}{3x^2y^4}$

g) $\frac{x^3y^3 \cdot x^3}{4x^2}$

h) $\frac{3x^2y^2}{2x^{-1} \cdot 4yx^2}$

i) $\frac{x}{(2x^0)^2}$

j) $\frac{2m^{-4}}{(2m^{-4})^3}$